SYNTHETIC DATA — 교육용 가상 데이터이며 실제 제품 사양·운영 기준과 무관합니다.

ESS BMS Controller V3 (ESS-BMS-V3) — Datasheet

1. Overview

Rack-level battery management system controller for LFP racks. Product family: Energy storage systems. Lifecycle status: **ACTIVE**.

2. Technical specifications

2.1. Ratings

Item	Value
Cell chemistry	LFP
Cells monitored per rack	Up to 240 cells in series

2.2. Performance and characteristics

Item	Value
Cell voltage accuracy	±5 mV
SOC estimation accuracy	±3%

2.3. Endurance and communication

Item	Value
Communication	CAN 2.0B (PCS), Modbus TCP (EMS)

2.4. Protection and options

Item	Value
Cell over-voltage protection (default)	3.65 V
Cell over-temperature protection (default)	55 °C

3. Installation data

3.1. Terminals

- Dimensions (W x H x D): 482 x 132 x 400 (19" 3U) mm
- Weight: 6 kg
- Degree of protection: IP20
- Operating temperature: -10 to +50 °C

4. Standards and certifications

Applicable standards:

- IEC 62619
- UL 1973 (component)

Certifications held:

- CE
- KC

Certificates and declarations of conformity are available from the Certification & Compliance team on request. Certifications not listed above are not held for this product; project-specific evaluation can be requested.

5. Ordering information

Item	Value
Ordering code	ESS-BMS-V3
Standard lead time	8 weeks ex-works
Minimum order quantity	1
Warranty	36 months (see POL-WARRANTY)
Firmware	latest 3.50 (see RN-ESS-BMS-V3)

6. Related documents

- IG-ESS-BMS-V3 — Installation and Commissioning Guide
- TG-ESS-BMS-V3 — Troubleshooting and Error Code Guide
- RN-ESS-BMS-V3 — Firmware Release Notes
- FAQ-ESS — Energy storage systems FAQ

SYNTHETIC DATA — 교육용 가상 데이터이며 실제 제품 사양·운영 기준과 무관합니다.

ESS BMS Controller V3 (ESS-BMS-V3) — Installation and Commissioning Guide

1. Safety

Installation, wiring and commissioning must be performed by qualified personnel. Disconnect and lock out all power sources before work. Verify absence of voltage.

2. Receiving and storage

Inspect the packaging on arrival and report damage within 7 days (see POL-RETURN). Store indoors in a dry place between -25 and +55 °C, relative humidity below 85% without condensation.

3. Mounting and clearances

- Required clearances: 1,000 mm in front for maintenance; ambient controlled by HVAC between 15 and 30 °C
- Operating temperature: -10 to +50 °C
- Degree of protection: IP20
- Above 1,000 m altitude, apply derating per the datasheet.

4. Wiring and tightening torques

- Terminal tightening torque: see dimension drawing
- Use copper conductors rated for at least 75 °C.
- Separate power and signal cables by at least 200 mm.

5. Commissioning procedure

1. Check battery rack insulation resistance
2. Verify CAN communication between PCS and BMS
3. Select grid code profile (parameter G-01)
4. Perform pre-charge test
5. Run charge/discharge test at 20% power

6. Pre-energisation checklist

☐ Product rating matches the drawing
☐ All terminals tightened to specified torque
☐ Protective earth connected
☐ Insulation resistance measured and recorded
☐ Settings / parameters recorded

7. Common commissioning issues

Symptom	Likely cause	See
DC insulation fault	Moisture in battery rack DC cabling	TG-ESS-BMS-V3 §3.1
BMS communication loss on PCS	Firmware compatibility: BMS 3.4 heartbeat vs PCS < 2.62	TG-ESS-BMS-V3 §3.2
Intermittent CAN errors	CAN termination missing	TG-ESS-BMS-V3 §3.3
Cell over-temperature	Container HVAC failure	TG-ESS-BMS-V3 §3.5

On-site commissioning support can be requested through the Global CS Center. For energy storage systems the request is handled by the technical support team.

SYNTHETIC DATA — 교육용 가상 데이터이며 실제 제품 사양·운영 기준과 무관합니다.

ESS BMS Controller V3 (ESS-BMS-V3) — Firmware Release Notes

1. Summary

Latest firmware: **3.50**. Updating to the latest version is recommended for all units.

2. Version history

2.1. Version 3.50 (2026-01-19)

- Fixed: SOC jump of up to 15% after long rest period
- Fixed: contactor status mismatch alarm on cold start

2.2. Version 3.40 (2025-05-19)

- Changed: CAN heartbeat period from 100 ms to 200 ms
- Improved: passive balancing threshold 30 mV → 20 mV

2.3. Version 3.30 (2024-10-28)

- Baseline release

3. Update procedure

1. Back up parameters / project before updating.
2. Connect the engineering tool (v4.6 or later).
3. Do not switch off power during the update.
4. After the update, verify the version and restore parameters if required.
5. If the update fails, do not power-cycle the unit repeatedly; contact support for the recovery procedure.

4. Compatibility

PCS firmware 2.62 or later is required with BMS firmware 3.40 or later (CAN heartbeat change).